QP CODE: 24900203

Reg No:.....

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First Semester

Discipline Specific Core Course - MG1DSCPHY100 - FOUNDATIONS OF PHYSICS

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A
Short Answer Type Questions
Answer any 7 questions
Each qusetion carries 2 marks

1	Convert a) 10 nanometres (nm) to meters (m). (b) 10 meters (m) to micrometres (μm).	[U]	[1]
2	Two vectors A and B are Parallel. What will be their vector product and scalar product?	[A]	[1]
3	Differentiate between average acceleration and instantaneous acceleration.	[U]	[2]
4	What is the horizontal component of velocity in projectile motion?	[U]	[3]
5	How is acceleration related to velocity of a particle? Will the acceleration of a particle become zero when it moves with a constant speed along a curved path?	[U]	[3]
6	A car is initially moving in reverse. The driver applies brakes, slowing the car. What is the direction of car's acceleration, relative to the car?	[An]	[3]
7	What is the power delivered when work of 1000 J is done in 2 seconds	[A]	[4]

8	What is the difference between positive work and negative work?	[U]	[4]
9	Assume that the earth moves around the sun in a circular orbit. Does the sun do any work on the earth?	[An]	[4]
10	Write the syntax of if else statements in Python.	[K]	[5]

(7 × 2 = 14)

Part B

Short Essay Type Questions
Answer any 4 questions
Each question carries 6 marks

11	Sketch and show the resultant R of vector sum of A and B.	[U]	[1]
12	A cheetah is found at 20 m to the east of a vehicle. At time t = 0, it begins to run due east towards its prey which is at 50 m to the east of the vehicle. During the first 2.0 s of the chase , the Cheetah's x- coordinate varies with the time according to the equation $x = 20m + (5m/s^2)t^2$. Find the Cheetah's instantaneous velocity at t1 = 1 s by taking, $\Delta t = 0.1$ s, then 0.01 s and 0.001 s.	[A]	[2]
13	An iceboat is at rest on a frictionless horizontal surface. Due to the blowing wind, 4.0 s after the iceboat is released, it is moving to the right at 6.0 m/s (about 22 km/h, or 13 mi/h). What constant horizontal force does the wind exert on the iceboat?	[A]	[3]
14	Discuss the transfer of energy in trampoline jumping.	[An]	[4]
15	What is the work done by a constant force?	[U]	[4]
16	Write a Python program to determine if a given number is positive, negative, or zero.	[A]	[5]

(4 × 6 = 24)

Part C

Essay Type Questions
Answer any 1 question
Each question carries 12 marks

17	Explain the operators in Python and their rules of precedence.	[U]	[4]
18	Give a detailed account on errors in python programming.	[K]	[5]

(1 × 12 = 12)

END OF THE QUESTION PAPER

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